

Direct and Alternative current:

Most of the examples dealt with so far, and particularly those utilizing batteries, have constant voltage sources. Once the current is established, it is thus also a constant. *Direct current (DC) is the flow of electric charge in only one direction. It is the steady state of a constant-voltage circuit.*

Most well-known applications, however, use a time-varying voltage source. *Alternating current (AC) is the flow of electric charge that periodically reverses direction.* If the source varies periodically, particularly sinusoidally, the circuit is known as an alternating current circuit. Examples include the commercial and residential power that serves so many of our needs. Figure 1 shows graphs of voltage and current versus time for typical DC and AC power. The AC voltages and frequencies commonly used in homes and businesses vary around the world.

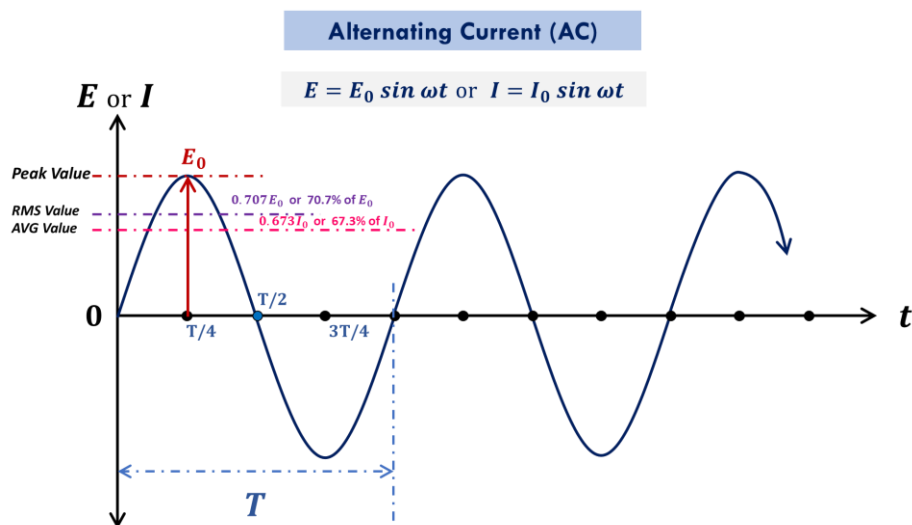
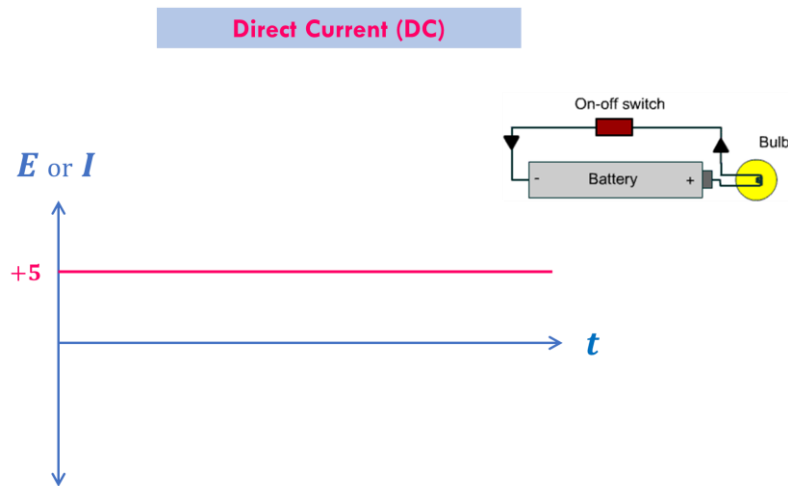


Figure 2 shows a schematic of a simple circuit with an AC voltage source. The voltage between the terminals fluctuates as shown, with the *AC voltage* given by

$$V = V_0 \sin \omega t$$

where V is the voltage at time t and V_0 is the peak voltage. For this simple resistance circuit, $I=V/R$, and so the *AC current* is

$$I = I_0 \sin \omega t$$

Common Formula of AC:

- Peak value, V_0 or E_0 and I_0
- Angular velocity or angular frequency, $\omega = 2\pi f$ or $\omega = \frac{2\pi}{T}$
- Frequency, $f = \frac{1}{T}$ or $f = \frac{\omega}{2\pi}$
- Time period, $T = \frac{1}{f}$ or $T = \frac{2\pi}{\omega}$
- Average value for full cycle, $I_{avg} = 0$ or $E_{avg} = 0$
- Average value for half cycle, $I_{avg} = 0.637I_0$ or $E_{avg} = 0.637E_0$
- Root Mean Square (RMS) value for full cycle, $I_{rms} = 0.707I_0$ or $E_{rms} = 0.707E_0$
- Form Factor = $\frac{rms \text{ value}}{avg \text{ value}} = 1.11$